

# Andre C.A. de Carvalho

Full-Stack Developer · Go · TypeScript · Python · Flutter

Uberlândia, MG, Brazil · Open to remote, hybrid, or on-site  
and2carvalho@gmail.com · linkedin.com/in/and2carvalho · github.com/and2carvalho

## SUMMARY

Full-Stack Developer with 7+ years shipping web and mobile products to production: Go microservices (gRPC, NATS), React/Next.js frontends, Flutter apps, and marketplace and payment integrations. Hands-on applied AI in production: a Vision LLM-backed OCR pipeline that raised accuracy from 28% to 75%. Strong data background (ETL, SQL, Python). Currently pursuing an M.Sc. in Computer Science at UFU, researching systems observability and verification of computational behavior.

## SKILLS

|                   |                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------|
| Applied AI        | Vision LLM via API · OCR pipeline · Ollama · MCP · AI artifact provenance                       |
| Backend           | Go · Python · Node.js · gRPC · REST · NATS JetStream · Microservices · Outbox Pattern · KrakenD |
| Frontend & Mobile | React · Next.js · TypeScript · React Query · Zustand · Storybook · Flutter · React Native       |
| Data & Infra      | PostgreSQL · MongoDB · Redis · ElasticSearch · Docker · Kubernetes · ETL/SQL · Power BI         |
| Integrations      | Mercado Livre · Amazon · Shopee · PagarMe · PinPag · Eventim                                    |
| Quality           | Jest · Vitest · Playwright · CI/CD                                                              |

## EXPERIENCE

Full-Stack Engineer · Beezoo Labs Jan 2025 – Present

Multi-tenant SaaS platform for loyalty and digital commerce (malls & retail)

- Built from scratch the **Showcase** microservice (digital storefronts with QR codes, pricing policies, and catalog enrichment via NATS events) plus its mobile-first frontend — end-to-end feature: .proto → Go service → admin panel → public app.
- Led the **app-whitelabel** initiative in the Go microservices monorepo (gRPC, NATS JetStream, Outbox Pattern, PostgreSQL, KrakenD), enabling multi-tier mobile app configuration per tenant.
- Developed white-label **Flutter** apps for consumers and partners — loyalty, campaigns, Eventim integration, biometric login, and a unified design system.
- Solo-built an **OCR** service for Brazilian tax receipts (NF-e/NFC-e) in Python/gRPC, chaining QR → Tesseract → PaddleOCR → Vision LLM (via API); raised accuracy from **28% to 75%** on a 1,138-image real-world dataset; deployed on Docker + Kubernetes.

Stack: Go · Python · TypeScript · Dart · gRPC · NATS · PostgreSQL · Next.js · React · Flutter · Docker · Kubernetes

Software Engineer · Hubsell May 2022 – Dec 2024

E-commerce integration platform

- Built frontend and integrations for the **Seller Center**, the interface sellers use to run their sales on Mercado Livre, Amazon, Shopee, and other marketplaces.
- Built the **POS (Point of Sale)** and **Shopping Assistant** features, aimed at seller conversion.
- Integrated **payment gateways** (PagarMe, PinPag) into the seller dashboard.
- Tackled loading bottlenecks on heavy screens (dashboards, sales panels) with **ElasticSearch**-backed search and client-side data caching via **React Query**.
- Shipped a mobile app for Android and iOS with **Expo**.

Stack: React · Next.js · Node.js · MongoDB · ElasticSearch · Zustand · React Query · Expo · Firebase

3Ponto — financial management platform (mobile app + web dashboard)

- Worked across the full product lifecycle — mobile app (**React Native**) and web dashboard (**React**) — from translating business requirements to shipping in production.
- Developed backend services in a **microservices** architecture (Node.js/Express, PostgreSQL).
- Led migrations and integrations with legacy systems.
- Led frontend work: feature delivery, bug fixing, and performance tuning.

Stack: React · React Native · Node.js · Express · PostgreSQL · Microservices

## Freelance Consultant — Data & Business Systems · Upwork

2016 – 2019

- Built **ETL** pipelines and **Power BI** dashboards for help desk and operations analytics.
- Implemented an **ERP** (Odoo/OpenERP) with custom business workflows and automations.
- Automated data models with **SQL** and **Python** for executive dashboards.

## EDUCATION & RESEARCH

### M.Sc. in Computer Science · Federal University of Uberlândia (PPGCO/UFU)

2026 – in progress

- **Research:** minimal representations of execution traces for systems observability — compact memory-access signatures that flag performance degradation (e.g.,  $O(n \log n) \rightarrow O(n^2)$ ) without per-class calibration, with an explicit criterion for when the representation stops being sufficient.
- **Method:** pre-registered hypotheses and reproducible experiments (TypeScript + Python); paper series in preparation.
- **Published artifact:** ( $D, \delta$ ): a two-feature signature for memory-access traces — code, data, pre-registered validation and robustness analysis, reproduced in CI. DOI [10.5281/zenodo.23198655](https://doi.org/10.5281/zenodo.23198655) · [github.com/and2carvalho/memory-access-signature](https://github.com/and2carvalho/memory-access-signature).
- **Research tooling — SEIF Protocol** ([seifprotocol.com](https://seifprotocol.com)): cryptographic provenance (Ed25519, time-anchoring via OpenTimestamps) and auditable governance for AI-assisted work, applied to the research itself. Public prototypes: [Carimbo](#) and [Vigília](#).

### B.A. in Business Administration · União de Faculdades Metropolitanas de Maringá

2016

## LANGUAGES

Portuguese

Native

English

C2 Proficient — EF SET certificate ([cert.efset.org/en/Eukv37](https://cert.efset.org/en/Eukv37))